SAKSHAM CHAWLA

Software Engineer

CONTACT

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CLEARANCE

TOP SECRET
06/2024 – Current
U.S. Citizen

EDUCATION

MASTER'S DEGREE,
MACHINE LEARNING
Georgia Institute of Technology
2024 — Current

BACHELOR'S DEGREE,
COMPUTER SCIENCE
Virginia Tech
2020 — 2023
GPA: 3.81; Magna Cum Laude

SKILLS

| Languages

- Python
- Java
- C/C++
- SQL
- JavaScript
- HTML/CSS
- R
- VBA

| Frameworks

- TensorFlow, Keras, PyTorch
- Scikit, Scipy
- Numpy, Pandas, Matplotlib
- NodeJS, React
- IaaS, PaaS, Saas

SUMMARY

Software Engineer with strong fullstack development skills and famialiar with Agile delivery. Passionate about delivering solutions with highest quality and ability to use the cloud capabilities that drive higher resiliency with a focus on user experience. Love learning new tech and always looking for opportunities to experiment and apply my knowledge in new projects.

Team player and assiting and motivating software engineers guiding them with my technical insights and building strong teams. My strength is driving alignment between business needs, technology teams, and product organization. Enjoyed having the opportunity to manage a team of junior developers.

I am passionate about the use and adoption of artificial intelligence and machine learning, with a goal of use of this capability in everyday solution delivery and currently sharpening my skills through advanced coursework in my master's program at Georgia Tech, aiming to leverage these skills to drive innovative solutions using LLM, AI and Agentic AI solutions.

WORK EXPERIENCE

SOFTWARE ENGINEER
BCubed Engineering | Aug 2023 — Dec 2024

US Navy Experience
Software Engineer, working on a project for the US Navy on the Mobile User Objective System (MUOS) team, responsible for developing software in efforts of revolutionizing secure ultra-high frequency satellite communications for mobile forces. Working in collaboration with Digital Signal Processing teams and Systems Engineering teams.

- Led the design and development of foundation set of Global API's for an analog-to-digitial signal converter for the U.S. Navy MUOS team, optimizing enhanced radio connectivity for end users impacting 1+ million U.S. military warfighters.
- Designed, developed, and deployed Python Kopf operators to automate updates to appliances running on FPGA boards.
- Administered deployment process (argoCD) of software, production infrastructure (k8s), image hub (Docker), monitoring/alerting systems (Prometheus), and messaging bus system (NATS Jetstream), providing a stable application experience. Utilized GitLab for version control and implemented Kanban for workflow management, optimizing collaboration and deployment efficiency.
- Worked with Digitial Signal Processing and Systems Engineering teams to create and test Verification and Validation (V&V) Procedures, aiming to align software components with key government requirements. Used automated QA testing and containerized Python and Bash Scripts to automate verification tests, enhancing efficiency and accuracy in testing processes.
- Managed a team of 3 Junior developers providing guidance in areas including Kubernetes, Python, and version control. Faciltated their learning through code reviews, hands-on-support, and project management, ensuring effective integration into team workflows, while also contributing to their professional development.

S A K S H A M C H A W L A

S o f t w a r e E n g i n e e r

SKILLS

| Tools

- Kubernetes
- Docker
- Git/Version Control, CI/CD
- GitHub Co-Pilot
- LaTex
- PowerBI
- Tableau
- NATS
- Prometheus
- MongoDB
- argoCD
- UML Structure

| Platforms

- Linux
- AWS
- Azure
- Firebase
- VS Code
- PyCharm
- Eclipse
- Lens (Kubernetes)

| Soft Skills

- Product Innovation
- Leadership
- Event Management
- Public Speaking
- Adaptability
- Flexibility
- Time Management
- Communication

WORK EXPERIENCE CONTINUED

UNDERGRADUATE TEACHING ASSISTANT

Virginia Tech Dept. of Computer Science | Jan 2023 — May 2023

Head Undergraduate Teaching Assistant (UTA) for Machine Learning

- Teaching Machine Learning for CS 4824 and ECE 4424 to 55 Computer Science students and 35 Computer Engineering students focusing on concpets of supervised and unsupervised learning.
- Led and instructed numerous lectures discussing concepts such as K-Nearest Neighbor, Baysian Statistics, and Nueral Networks.
- Developed a PyTorch Jupyter Notebook tutorial focused on concepts of Nueral Networks and provided a live demo class lecture for both sections.
- Led weekly office hours sessions focused on helping students with debugging, understanding core concepts, and providing project assistance.

SOFTWARE ENGINEER INTERN

Aptive Resources | Aug 2021 — Oct 2022

Software Engineer Intern for the Internal R&D Division. Focused on concepts/ideas translation into prototypes in the areas of web application development, AI/ML use cases, and Data Analytics

- Developed a full-stack onboarding application using React.js, Node.js, and Python to streamline the employee onboarding process.
- Crafted a technical machine learning proposal that won Phase 1 of the Navy's Controlled Unclassified Information (CUI) Tool Automation Challenge.
- Developed a VBA macro that scans large documents to generate a table of contents by extracting acronyms and their definitions, intended for use by 500+ company employees for efficient document navigation.

PROJECTS

Optimized Stock Trader

Aug 2024 — Dec 2024

Developed an ensemble learning model leveraging Random Forest and feed-forward cross-validation to optimize stock market trades. Analyzed adjusted close prices for 50+ tickers and incorporated technical indicators like Bollinger Bands, MACD, EMA, and SMA as features. Implemented a lookahead technique to define target variables (holding, long, or short positions). Achieved performance surpassing multiple benchmarks, demonstrating superior normalized portfolio values and cumulative returns.

COVID-19 Analysis

Oct 2021 — Dec 2021

Collected and anlayzed a dataset of 150,000+ COVID-19 data points from the World Health Organization (WHO), spanning from December 31, 2019 to October 22, 2021. Used a form of unsupervised learning by implementing K-Means Clustering and the Elbow Method to identify countries requiring the most medical attention during the pandemic. Applied regression analysis to forecast future cases.